Clinical Decision Support System

MRI Brain Tumor Analysis Report

Patient ID	MRI-4_62F94C
Name	Ahmed Bensalem
Age	38 years
Sex	Male
Comorbidities	diabetes, hypertension
Symptoms	headache, seizure, weakness
Date/Time	2026-05-21T08:05:39.922465
Run ID	20260521_080504_62f94c
Tumor Type	GLIOMA
Confidence	100.0%
Radiologist	Approved

Clinical Summary

Glioma detected. This is a primary brain tumor arising from glial cells. High-grade lesions are fast-growing and require urgent neurosurgical assessment and likely multimodal treatment (surgery, radiation, chemotherapy).

Patient-Specific Clinical Considerations

• Young patient (<40 yrs): IDH-mutant glioma is more likely — favourable prognosis. Molecular profiling (IDH, 1p/19q, MGMT) is essential.
• Diabetes mellitus: corticosteroid use (e.g., dexamethasone for cerebral oedema) will cause significant hyperglycaemia. Close glucose monitoring and insulin dose adjustment are required.
• Hypertension: maintain strict blood-pressure control peri-operatively (target <140/90 mmHg). Uncontrolled hypertension increases intracranial haemorrhage risk and may confound MRI perfusion sequences.
• Seizure presentation: initiate anti-epileptic therapy — levetiracetam (Keppra) is preferred (no hepatic enzyme induction, no interaction with chemotherapy). EEG and neurology review recommended.
• Headache: assess for raised intracranial pressure (papilloedema, morning predominance, positional worsening). Consider dexamethasone if significant perilesional oedema is present on imaging.
• Motor weakness: document laterality and onset. Urgent physiotherapy referral and falls-risk assessment. Consider corticosteroids if acute deterioration due to oedema.

Recommended Next Steps

1. URGENT neurosurgery consultation
2. MRI with gadolinium contrast + DTI/PWI for surgical planning
3. Neuro-oncology consultation for adjuvant therapy planning
4. Molecular profiling (IDH, MGMT) for prognosis
5. Consider enrollment in clinical trials

Clinical Guidelines

GLIOMA

Definition: Glioma is a type of tumor that starts in the glial cells of the brain or spine. Glial cells support and protect neurons.
Characteristics:
- Fast-growing, often malignant tumors
- Can originate anywhere in the brain or spinal cord
- Most common primary brain tumors in adults (approximately 30% of all brain tumors)
- Graded I-IV based on aggressiveness (WHO classification); Grade IV (glioblastoma) is the most aggressive
- Often infiltrative, making complete surgical resection difficult

Typical Imaging Findings (MRI):
- Irregular, poorly defined borders
- Heterogeneous signal intensity on T1/T2 sequences
- Ring enhancement with contrast in high-grade gliomas
- Surrounding vasogenic edema (T2/FLAIR hyperintensity)
- Mass effect with midline shift in large tumors
- Central necrosis in glioblastoma

Recommended Next Steps:
1. Confirm diagnosis with stereotactic biopsy if resection is not feasible
2. MRI with and without gadolinium contrast for full staging
3. Advanced MRI sequences: DTI (diffusion tensor imaging) for white matter tracts, PWI (perfusion-weighted imaging) for vascularity
4. URGENT referral to neurosurgery for surgical assessment and possible resection
5. Neuro-oncology consultation for adjuvant chemoradiation planning
6. Consider molecular profiling (IDH mutation, MGMT methylation) for prognosis and treatment
7. Enrollment in clinical trials if available

Prognosis:
- Depends heavily on WHO grade, molecular markers, and patient age
- Grade II: median survival 5-15 years
- Grade III: median survival 2-5 years
- Grade IV (glioblastoma): median survival 14-16 months with standard Stupp protocol (surgery + TMZ + radiation)
- IDH-mutant tumors carry better prognosis than IDH-wildtype

Treatment:
- Maximal safe surgical resection (if feasible)
- Fractionated external beam radiation therapy (60 Gy in 30 fractions for GBM)
- Temozolomide (TMZ) chemotherapy concurrent with and adjuvant to radiation
- Tumor treating fields (TTFields) for recurre...

This report is generated by an AI-assisted clinical decision-support system. It is NOT a substitute for professional medical diagnosis. All findings must be confirmed by a qualified radiologist or physician.